

NanoMatch — Submission Cover Letter

Project: NanoMatch — Ultra-Low Latency Order Matching Engine **Author:** Shaurya Atram
Submission date: 2026-05-22 **Addressed to:** Shubham Rane, Tanishq Kothari (FEC IIT Guwahati DIY '26 mentors)

Overview

NanoMatch is a from-scratch C++20 limit-order-book matching engine built for the FEC IIT Guwahati DIY '26 project. It implements strict price-time priority, integer-tick pricing, and a fully cache-resident hot path with zero heap allocation, zero locking, and zero I/O on the matcher thread. The engine ingests NASDAQ TotalView-ITCH 5.0 data via memory-mapped I/O and outputs matched trades through a lock-free SPSC ring to an asynchronous logger.

Headline Performance Numbers

Microbenchmark results from `nanomatch_bench 10000000` on Intel i7-14650HX, GCC 13.3.0, `-O3 -march=native -DNDEBUG` (full table in docs/REPORT.md, Section 4):

Metric	Naive baseline	Optimized engine	Improvement
Throughput	4.64 M ops/s	17.93 M ops/s	3.9x
Median latency (p50)	96 ns	24 ns	4x
Max latency	137 ms	148 us	925x

Real-Data Validation

The optimized engine processed the complete NASDAQ TotalView-ITCH 5.0 feed for 2019-07-30 (8.66 GB, 282 million messages). It matched 105,722,822 trades in 35.5 minutes wall-clock time with zero logger drops. The output covers the full 16-hour session from pre-market open (04:00 EST) through after-hours close (20:00 EST).

Where to Look

Deliverable	File
Project overview, build/run instructions	README.md
Performance Benchmark Report	docs/REPORT.md
CPU flame graph (naive vs optimized)	docs/flame_combined.svg
Hardware and toolchain specs	docs/hardware.md

Caveats

The engine is single-instrument and in-process; there is no PCAP network layer, and hardware cache-miss PMU counters are unavailable under WSL2 (flame graph attribution uses DWARF software stack sampling instead).

Acknowledgement

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PDF version: regenerate with `pandoc docs/REPORT.md -o docs/REPORT.pdf --pdf-engine=xelatex -V geometry:margin=1in -V fontsize=11pt --toc` once pandoc is installed.